



DECEMBER 2024
THERMAL ENGINEERING-I

3247848

Time Allowed: 3 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and

1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): $2 \times 10 = 20$

- i. Cochran boiler is a
(a) horizontal fire-tube boiler, (b) horizontal water-tube boiler,
(c) vertical water-tube boiler, (d) vertical fire tube boiler.
- ii. With increase of pressure, latent heat of evaporation
(a) increases, (b) decreases, (c) remain constant (d) may increases or decreases
- iii. Superheated vapour behaves
(a) Exactly as gas, (b) as steam, (c) as ordinary vapour, (d) approximately as gas.
- iv. Measurement of temperature is based on _____ law of thermodynamics.
(a) zeroth, (b) first, (c) second, (d) third.

- v. Boyle's law relates to
(a) constant volume, (b) constant pressure, (c) constant temperature, (d) none of these.
- vi. Water tube boilers are those in which
(a) flue gases pass through tubes and water around it, (b) water passes through the tubes and flue gases around it, (c) work is done during adiabatic expansion, (d) change in enthalpy.
- vii. The enthalpy of a substance is defined as
(a) $H = U - PV$, (b) $H = U + PV$, (c) $H = -U + PV$, (d) $H = -U - PV$.
- viii. When water is heated with the rise of temperature, it consumes
(a) latent heat, (b) sensible heat, (c) both (a) and (b) (d) none of these.
- ix. Heat is
(a) intensive properties, (b) extensive properties, (c) both intensive and extensive properties, (d) can't say



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- x. The temperature at which the Celsius and Fahrenheit scales read the same is
(a) 60° (b) -60° (c) -40° (d) 40°
- xi. The maximum heat loss in a boiler occurs due to
(a) moisture in fuel, (b) dry flue gases, (c) steam formation, (d) unburnt carbon.
- xii. Dryness fraction(x) of superheated steam is
(a) equal to 0, (b) greater than 1, (c) less than 1, (d) equal to 1.
- xiii. The function of Blow down valve of a boiler is
(a) to remove sediments (b) to build sediments (c) to remove flue gas (d) to remove ash
- xiv. The natural draught in boiler is produced by
(a) steam jet (b) centrifugal fan (c) chimney (d) both (a) and (b)
- xv. An economiser is installed in a boiler primarily to -
(a) superheat the steam (b) reduce fuel consumption
(c) increase steam pressure (d) all of these

GROUP-B

2. a) State First law of thermodynamics. What are the limitations of this law?
b) Deduce the steady flow energy equation. [3+5]
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3. a) Define with brief explanation: PMM-I and PMM-II
b) Define the following with example: (i) system, surroundings and boundary, (ii) intensive and extensive properties [4+4]
4. a) State Boyle's law and Charle's laws.
b) Show that $C_p - C_v = R$. [3+5]
5. a) What is universal gas constant?
b) 0.2 m^3 of an ideal gas at a pressure of 2 MPa and 600 K is expanded isothermally to 5 times the initial volume. It is then cooled to 300 K at constant volume and then compressed back polytropically to its initial state. Determine the net work done during the cycle. [2+6]
6. a) Deduce the expression for work done during adiabatic process.
b) Write shortly on (i) Dryness fraction of steam (ii) Air preheater of boiler [4+(2+2)]



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7. a) What is meant by critical point? Draw the outline of Mollier chart and show critical point on it.
b) How much heat is needed to convert 5kg of water at 45°C into steam at pressure of 4 bar and 80 per cent dry? Take specific heat of water as 4.187 kJ/kg-K .
[(2+2)+4]
8. a) Differentiate between fire tube boiler and water tube boiler.
b) What is feed water? What are the necessities of boiler feed water treatment?
c) Define equivalent evaporation.
[3+(1+2)+2]
9. a) Explain with neat sketch the construction and working principle of Cochran boiler.
b) A boiler uses 18 kg air per kg of fuel. The mean temperature of chimney gases is 315°C and that of outside air 27°C . Determine the minimum height of chimney required to produce a draught of 20 m.
[5+3]

